

Skin-Contact Materials in NPWT Systems:

Navigating Adhesion, Conformability, and Interface Performance

A practical guide for medical device engineers and OEM product teams

If you've spent time developing a Negative Pressure Wound Therapy (NPWT) system, you already know the central headache: the same interface that has to seal reliably under continuous negative pressure is the one that gives you the most difficult material trade-offs. Adhesion, conformability, atraumatic removal, and moisture management all want different things from the same component. You can't optimize for all of them at once.

That tension doesn't go away but it can be managed if you understand where the real trade-offs live and what the failure modes look like before you hit design verification.

This paper focuses on three areas where early specification decisions have an outsized impact on how a program unfolds: adhesive system selection, foam scaffold specification, and drape film performance. These aren't the only places where NPWT development programs run into trouble, but they're the most consistent ones and the ones where getting things right early saves the most time and cost downstream.



Adhesive System Selection: Finding the Right Bond-Force Window

Choosing an adhesive for an NPWT drape sounds straightforward. In practice, it's one of the more nuanced specification decisions in the program.

Acrylic pressure-sensitive adhesives are the historical default in wound care, and for good reason. They're well-characterized, they bond reliably across a wide range of substrates, and there's a well-worn path through ISO 10993 biocompatibility testing. For many NPWT applications, they're a perfectly reasonable choice.

The challenge is the removal-force side of the equation. Acrylic adhesives can exhibit adhesion buildup over extended wear periods, which means the peel force at removal may be meaningfully higher than what you measured at application, sometimes enough to push you outside your design window. For programs targeting high-risk patient populations or extended wear durations, that's not a minor comfort consideration. It's a specification problem.

Silicone-based adhesives handle this differently. Their adhesion mechanism produces a more controlled and consistent peel-force profile over time. You tend to get a softer, more predictable release that holds up better across extended wear. The trade-off is seal robustness under contaminated conditions. Silicone adhesives are more sensitive to surface energy variation. Moisture, residual topical agents, and exudate at the

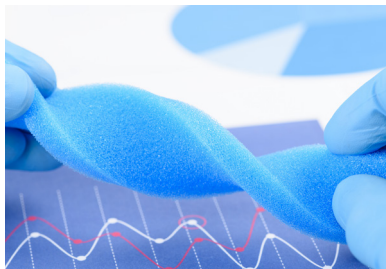
bond line can all degrade seal integrity in ways that are harder to predict from supplier data sheets alone. That doesn't make silicone the wrong choice, it means the characterization work needs to go deeper.

One failure mode that catches programs off guard regardless of adhesive chemistry: lateral shear loading on articulating anatomy. A drape that passes seal integrity testing on a flat substrate under static conditions may fail on a mobile patient at the knee or shoulder. If your design inputs include high-motion anatomical sites, that loading condition needs to be in your verification testing not discovered in post-market surveillance.

Regulatory Note: ISO 10993-5 and 10993-10 are the baseline biocompatibility requirements for skin-contact adhesives. For extended-wear NPWT, continuous application beyond 72 hours, you'll also want dermal irritation data under ISO 10993-23 generated during design development, not assembled after design freeze. Adhesive reformulation post-510(k) submission is expensive and slow.

Key Takeaway: Nail down your wear-duration design inputs early, then choose your adhesive chemistry around the full peel-force envelope not just initial tack. And test your seal under the motion conditions the device will actually see in use.

Foam Scaffold Specification: Why Pore Architecture Is a Functional Parameter



The wound contact foam in an NPWT system isn't just a spacer. Its physical architecture, such as pore size, open-cell geometry, compressive stiffness, determines how pressure is distributed across the wound interface, how efficiently fluid moves to the canister, and how the material behaves mechanically over the wear period. Get the architecture wrong, and you'll see it in your system-level performance data.

Pore size is the variable that drives the most downstream consequences. Standard open-cell polyurethane foams for NPWT typically target 400-600 microns, or roughly 40-65 PPI, a range that balances fluid transport with adequate mechanical compliance under compression. Move outside that window in either direction and performance degrades in predictable ways. Pores that are too large produce non-uniform pressure distribution. Pores that are too small restrict fluid transport and reduce effective pressure transmission at the wound interface.

The manufacturing challenge is holding that specification at production scale. Polyurethane foam properties are sensitive to raw material lot variation, processing temperature and humidity, and equipment calibration. The temptation is to use compressive resistance as a proxy for foam quality in incoming inspection. It's easy to measure and it's correlated with pore structure. But it's not a sufficient indicator. Programs that rely on compressive resistance alone as their acceptance criterion tend to encounter surprises when they move from development samples to first production lots.

Foam specification drift between early development samples and first production lots is one of the most consistent sources of design verification failure we see across NPWT programs. Lock your supplier specifications and incoming inspection criteria before design freeze.

The right answer is to include pore size distribution as an explicit acceptance criterion, via image analysis or mercury intrusion porosimetry, rather than treating it as characterization data collected once and filed away. It adds cost to incoming inspection, but it's the right lever to pull when pore architecture is a functional performance parameter.

Polyvinyl alcohol foam is worth knowing about as an alternative. It's more hydrophilic, softer in compression, and tends to release more cleanly from the wound interface than polyurethane. The selection between the two isn't purely a performance decision. Regulatory precedent, manufacturing process compatibility, and total system cost all factor in. But it's worth evaluating both early if your design inputs leave the door open.

Key Takeaway: Treat foam pore architecture as a functional specification, not a characterization exercise. Build pore size distribution into your supplier qualification and incoming inspection criteria from the start.

Drape Film Selection: Balancing MVTR, Durability, and Conformability

Drape film selection is a three-variable trade-off problem, and the three variables don't move independently. Moisture Vapor Transmission Rate (MVTR), mechanical durability, and conformability are all functions of film thickness and base polymer. Improve one, and you typically give something up on at least one of the others.



MVTR governs how efficiently moisture vapor generated at the wound interface passes through the drape. When transmission is insufficient, moisture accumulates at the film-skin boundary, which degrades adhesive bond performance over the wear period, a failure mode that's independent of your adhesive chemistry. It's a system-level effect, and that's exactly where most MVTR specification errors originate.

The mistake we see most often: teams characterize MVTR at the film substrate level rather than through the complete drape-adhesive assembly. The adhesive coating can act as a partial vapor barrier, and a film that hits your MVTR target in isolation may fall short when tested as a bonded laminate. System-level MVTR testing through the complete assembly needs to be in your design verification protocol from day one.

On the durability side, higher MVTR generally means a thinner, more permeable film structure. That's good for vapor management and conformability, but it reduces resistance to tearing and puncture during patient handling. Thicker films hold up better mechanically but restrict vapor transmission and reduce drape conformability on contoured surfaces, which affects both seal integrity at the drape border and how the system performs during extended wear.

The right specification depends on your target application. A system designed for 48-hour acute inpatient use has a very different film specification than one targeting 7-day ambulatory wear on a high-motion anatomical site. Conformability, in particular, tends to be evaluated qualitatively during early development and quantitatively too late, after design freeze, as a verification activity rather than a design input. Build quantitative conformability characterization into your development protocol early.

Key Takeaway: Always test MVTR through the complete drape-adhesive assembly, not the film alone. And make sure your conformability requirements are defined and tested against realistic anatomical geometry under representative negative pressure loading before design freeze.

A Word on Biocompatibility: Don't Leave It for the End



For NPWT skin-contact components, biocompatibility evaluation under ISO 10993-1 requires chemical characterization data that traces back to raw material suppliers. The testing endpoints that matter most for extended-contact applications, including cytotoxicity, sensitization, irritation, and in some cases systemic toxicity, depend on formulation-level information that supplier data sheets rarely provide in the form FDA reviewers expect.

The gap between what suppliers know about their materials and what you need for a 510(k) submission is one of the most reliably avoidable schedule risks in class II device development. Suppliers are usually willing to share formulation detail under a confidentiality agreement, but you have to ask for it early, build it into your supplier qualification requirements, and give yourself time to iterate if the data has gaps. Teams that treat biocompatibility dossier assembly as a submission preparation task rather than a design development activity consistently underestimate how long it takes.

Key Takeaway: Build biocompatibility data collection into your supplier qualification process from the start. Don't wait until after design freeze to figure out what you need and what you're missing.

Putting It Together: A Systems Approach to NPWT Development



The three material domains covered here, adhesive system, foam scaffold, and drape film, interact with each other in ways that are genuinely difficult to predict from component-level testing alone. The places where NPWT programs tend to run into trouble aren't usually in any single component. They're at the interfaces: between the adhesive and the film, between the foam and the drape, between a design that looked solid at the bench and a use environment that's messier and more dynamic than the test setup.

The most effective way to surface those interaction effects early is system-level prototyping tested against realistic use conditions, not idealized bench configurations. It's not glamorous advice, but it's the most consistent differentiator between programs that move smoothly through design verification and those that don't.

At UFP Technologies, we work with OEM development teams across the full NPWT development cycle, from early materials selection and feasibility prototyping through design verification support and production scale-up. With locations in the U.S., Mexico, Dominican Republic, Costa Rica, Ireland, and Singapore, we support the global supply chain needs of major medical device OEMs through scalable manufacturing infrastructure, ISO 13485 certified quality systems, and FDA registered operations.

If you're working through adhesive, foam, or film trade-offs on an NPWT program or are trying to understand why something that looked right at the component level isn't holding up at the system level, we're happy to have a candid technical conversation about what we're seeing and where we might be able to help.

